

CALIFORNIA DEPARTMENT OF PARKS & RECREATION · NATURAL RESOURCES DIVISION

SCIENTIFIC RESEARCH & COLLECTION PERMIT — STUDY PROPOSAL

Diversity of coralline algae in northern California and their reproductive systems

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Abstract

Coralline red algae (Corallinales, Rhodophyta) are foundational to rocky intertidal and subtidal ecosystems, yet their diversity along the northern California coast remains poorly resolved and their reproductive phenology is largely undocumented. We propose a 12-month survey of coralline diversity across four state-park units spanning the North Coast Redwoods and Mendocino districts, pairing morphological voucher collection with DNA barcoding to build a verified regional inventory and to test whether the onset of tetrasporangial reproduction tracks a latitudinal gradient. Destructive sampling is held to the minimum required for determination; all collection sites, quantities, and curation practices are detailed below.

1 Objectives & hypotheses

The study has three objectives: (i) to compile a taxonomically verified inventory of articulated and crustose coralline algae across the study area; (ii) to characterize the reproductive phenology of the dominant articulated genus, *Corallina*, over a full annual cycle; and (iii) to evaluate whether reproductive timing and diversity differ between high- and low-visitor-impact sites.

We test two hypotheses. **H1:** coralline species richness is comparable between high- and low-impact sites of similar exposure, indicating resilience to visitor pressure. **H2:** the onset of tetrasporangial reproduction in *Corallina vancouveriensis* advances with decreasing latitude, consistent with a temperature-mediated phenological gradient.

2 Study area & collection sites

Field work will occur at four rocky intertidal sites within Del Norte Coast Redwoods, Prairie Creek Redwoods, and MacKerricher state parks. Sites were selected to pair a highly accessible, visitor-impacted shore with a comparable low-impact shore of similar wave exposure and substrate.

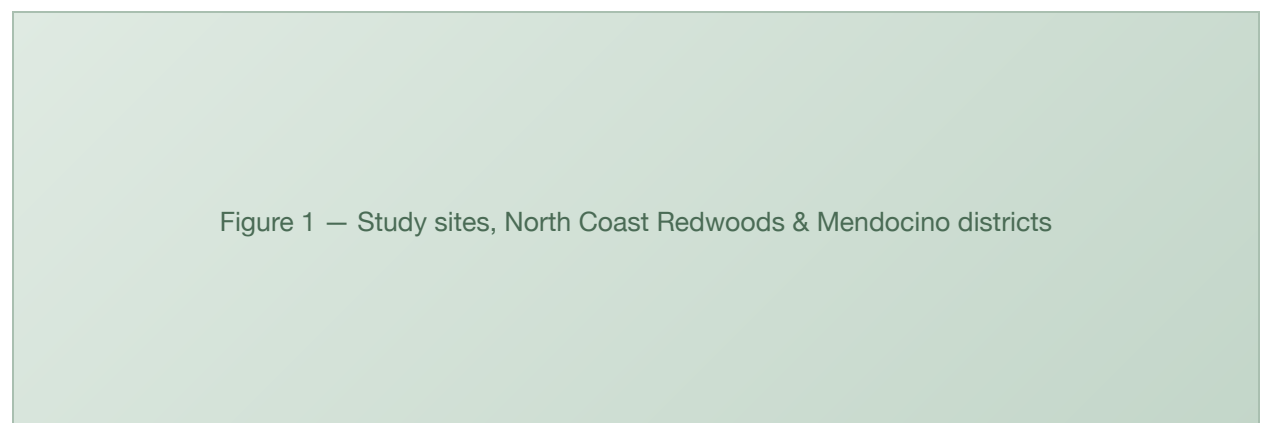


Figure 1 — Study sites, North Coast Redwoods & Mendocino districts

Figure 1. Four paired intertidal sites along the northern California coast. Full coordinates and access notes are provided in the accompanying study-area map (submitted separately).

3 Study procedures & methods

At each site we will conduct timed diversity surveys during spring low tides on a quarterly schedule (four visits per site over the permit year). Along three fixed 30-m shore-parallel transects per site, we will record percent cover of coralline taxa in 0.25 m² quadrats at 5-m intervals. Subtidal observations to –4 m MLLW will be made by AAUS scientific divers where conditions allow.

Voucher fronds will be collected by hand for morphological and reproductive determination and pressed for herbarium accession. A subsample of each morphospecies will be preserved in silica gel for DNA barcoding (*rbcL* and COI-5P) to confirm identifications against reference libraries. No power tools, chemicals, or substrate removal will be used.

4 Collection rationale & authorized specimens

Destructive sampling is limited to the minimum needed for morphological and reproductive determination. Most taxa require only a single frond; the expanded *Corallina vancouveriensis* allotment (50 fronds) supports the reproductive case study, which requires sampling across the annual cycle to capture the transition to tetrasporangial condition.

Species	Quantity	Portion	Condition
<i>Bossiella chilensis</i>	5	single frond	vegetative
<i>Calliarthron tuberculosum</i>	5	single frond	vegetative
<i>Corallina chilensis</i>	5	single frond	vegetative
<i>Corallina vancouveriensis</i>	50	25 single frond, 25 entire individual	reproductive & vegetative
<i>Lithophyllum impressum</i>	5	1 cm ²	vegetative
... 14 additional taxa	5 ea.	single frond / 1 cm ²	vegetative

Total authorized collection: 19 coralline taxa. All collection will occur outside designated marine reserves; the crew will remain outside the False Klamath Cove State Marine Reserve boundary at all times.

5 Data & data products

Occurrence records will be deposited to the Global Biodiversity Information Facility (GBIF) within six months of each field season. Vouchers will be accessioned at the Cal Poly Humboldt Vascular Plant & Algae Herbarium (HSC); DNA barcodes will be submitted to GenBank. A summary report will be provided to the Natural Resources Division within one year of permit issuance, and annually thereafter for the duration of any renewal.

6 Budget & personnel

The project is fully covered by a National Science Foundation systematics award (DEB-2043118); no state funds are requested. Field crews comprise the PI, one graduate researcher (subtidal survey lead and AAUS scientific diver), one laboratory technician, one undergraduate field assistant (field days only), and the herbarium curator. All personnel are employed or insured by the institution.

7 Literature cited

- Gabrielson, P.W., Hughey, J.R. & Diaz-Pulido, G. (2018). Genomics reveals abundant coralline diversity in the northeastern Pacific. *Journal of Phycology*, 54(2), 149–158.
- Hind, K.R. & Saunders, G.W. (2013). A molecular phylogenetic study of the tribe Corallineae (Corallinales, Rhodophyta). *Journal of Phycology*, 49(1), 103–114.
- Martone, P.T. (2010). Quantifying growth and calcium carbonate deposition in *Calliarthron* in the field. *Journal of Phycology*, 46(1), 12–21.
- Steller, D.L. & Cáceres-Martínez, C. (2009). Coralline algal rhodoliths as habitat. *Marine Ecology Progress Series*, 396, 49–60.
- (38 references cited; abbreviated for this application.)